

Motivating Students to Ask More Questions

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Abstract. Students don't ask enough questions in classrooms, in person or online. Asking questions has been found to be a critical skill toward developing critical thinking abilities, improve learning performance, as well as career development. However, 'how to ask productive questions' as a key skill, is not well studied. Therefore, the present paper introduces the question productivity index (QPI) and explores ways toward quantitatively and reliably measure student-generated questions.

Keywords: Question productivity · Critical thinking · Learning analytics

1 Background

1.1 Asking Questions Matters

It all starts with a good question! The popular saying is often heard in contexts of job interviews, formulating research questions, workplace discussions [1, 2], as well as classroom learning [3, 4]. Questioning, as a skill, has been found to have two major functions, one is learning, the other on liking. Recent research has shown that simply asking more questions can be an effective way to increase innovation, promote information exchanges, as well as building rapport and increase liking [1, 2]. This pattern was found in both in-person and online settings such as online chat rooms [11].

Despite various benefits of asking questions, research reveals that people do not ask enough questions or do not expect asking questions to be beneficial [2]. Not asking enough questions has been found to be a prevalent phenomenon in classrooms [3, 6–8]. Although question asking is generally encouraged and considered a sign of motivation, research showed that high-quality student-generated questions are scare and many students rarely ask any questions [4].

1.2 Benefits of Asking Questions Needs to Be Explained

Challenges of asking questions overshadow its benefits [1, 7–10]. Students may have different reasons and concerns not to ask questions, such as to avoid being seen incompetent [2, 12], cultural origins [5], language proficiencies [6]. By contrast, benefits of asking questions are rarely stressed in classrooms [6]. Teachers may

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welcome students to ask questions, yet skills and techniques of asking questions are rarely the center point of most classes. A vague understanding of why asking questions is beneficial, in itself, can prevent students from asking questions.

With the goal of helping students to ask more effective questions and increase learning performance, research is needed toward understanding (1) why asking more questions is good (2) what makes questions good.

Toward the above-mentioned goals, we proposed to develop a question productivity index and have expert teachers to rate students questions according to this index, with an aim to optimize neural nets that can automatically score student-generated questions.

2 Question Productivity Index (QPI)

As a pilot, expert college instructors were recruited to rate a set of student-generated questions from a diverse range of domains including planetary sciences, business, mathematics, etc. Each rater was asked to rate each question on its overall productivity and on the three individual dimensions of the QPI. Thereby, each rater scored each of 109 questions in four ways. Each question was rated by four raters, resulting in 16 scores per question. A QPI rubric were made available to raters to download as a PDF to help with their rating processes. The estimated rating time to complete the rating for 109 questions was around 2 h (Table 1).

Table 1. Descriptions of the QPI dimension

Name	Descriptions
Relevance	How relevant is the question to the larger learning goal?
Scale	The question takes the class one reasonable step from their current knowledge
Articulation	The question is well-posed and uses good grammar
Overall	An overall score on the productivity of the question

3 Early Findings and Discussions

A multiple regression model was conducted to investigate if the three dimensions of question productivity while controlling word count and course types. The results (see Table 2) indicated that three dimensions of QPI, together with word count and whether the question was raised in a STEM course, explained 82% of the variance.

It was found that all three dimensions, each significantly predicted overall productivity scores, Scale ($\beta = .15$, $p < .001$), Articulation ($\beta = .27$, $p < .001$), Relevance ($\beta = .62$, $p < .001$). Relevance was found to be the strongest predictor among the three.

Table 2. Multiple regression to predict average overall score

Predictor	*b*	*b* 95% CI	*β*	*beta* 95% CI	sr^2	sr^2 95% CI	*r*
(Intercept)	−0.41	[− 1.13, 0.32]					
Scale	0.14	[0.06, 0.22]	0.15	[0.06, 0.24]	.02	[− .00, .04]	.45**
Articulation	0.25	[0.13, 0.38]	0.27	[0.13, 0.40]	.03	[− .00, .06]	.79**
Relevance	0.54	[0.43, 0.65]	0.62	[0.49, 0.75]	.16	[.09, .24]	.87**
Word Count	0.00	[− 0.01, 0.01]	0.01	[−0.08, 0.10]	.00	[− .00, .00]	−.07
STEM	−0.03	[− 0.44, 0.38]	−0.01	[−0.09, 0.08]	.00	[− .00, .00]	.01

Note. A significant *b*-weight indicates the beta-weight and semi-partial correlation are also significant. *b* represents unstandardized regression weights. *beta* indicates the standardized regression weights. sr^2 represents the semi-partial correlation squared. *r* represents the zero-order correlation. *LL* and *UL* indicate the lower and upper limits of a confidence interval, respectively. *indicates $p < .05$. **indicates $p < .01$. $R^2 = .820$

Word count ($\beta = .01$, p = .82) and whether the question was raised in a STEM course ($\beta = -.01$, p = .87) did not show statistically significant relationship with the overall productivity score. This result is somewhat surprising since the content of question in a STEM subject can be different from those from a non-STEM subject, which may suggests that skills of asking questions may share more commonalities than differences. Further research is required to explain this null result (Fig. 1).

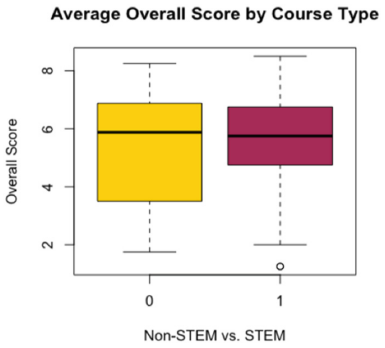


Fig. 1. Boxplot comparing average overall scores of questions between Non-STEM courses and STEM courses

4 Creating Feasible Classroom Intervention and Future Directions

Following up, we plan to collect more student-generated questions across a diverse range of subject areas and increase the number of expert raters. Student demographic information such as gender, age, and language background will also be incorporated in

the model to investigate if students belonging to any specific demographic groups tend to have distinctive questioning patterns.

In the process of optimizing the QPI index, one of the instructors in the research team has used the QPI rubric as a classroom questioning guide to encourage students to ask questions while being mindful of the various dimensions of the questions. Research has been planned to also measure students' goal orientations [12] and grit [13], as well as student grades to explore their relationships with question productivity.

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